

```
In [1]: # Import Libraries
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import classification_report, confusion_matrix
from sklearn.preprocessing import LabelEncoder

import joblib

# Training File Path
file_path = r"D:\Data\Prediction_Data.xlsx"

# Sheet Name
sheet_name = 'vw_churndata'

# Read Excel File
data = pd.read_excel(file_path, sheet_name=sheet_name)

# Clean Column Names
data.columns = data.columns.str.strip() \
                .str.replace(" ", "_") \
                .str.replace(r"^[^w]", "", regex=True)

# Show Data
print(data.head())
```

	Customer_ID	Gender	Age	Married	State	Number_of_Referrals	\
0	19877-DEL	Male	35	No	Delhi	7	
1	58353-MAH	Female	45	Yes	Maharashtra	14	
2	25063-WES	Male	51	No	West Bengal	4	
3	59787-KAR	Male	79	No	Karnataka	3	
4	28544-TAM	Female	80	No	Tamil Nadu	3	

	Tenure_in_Months	Value_Deal	Phone_Service	Multiple_Lines	...	\
0	27	NaN	Yes	No	...	
1	13	NaN	Yes	Yes	...	
2	35	Deal 5	Yes	No	...	
3	21	Deal 4	Yes	No	...	
4	8	NaN	Yes	No	...	

	Payment_Method	Monthly_Charge	Total_Charges	Total_Refunds	\
0	Credit Card	65.6	593.30	0.00	
1	Credit Card	-4.0	542.40	38.33	
2	Bank Withdrawal	73.9	280.85	0.00	
3	Bank Withdrawal	98.0	1237.85	0.00	
4	Credit Card	83.9	267.40	0.00	

	Total_Extra_Data_Charges	Total_Long_Distance_Charges	Total_Revenue	\
0	0	381.51	974.81	
1	10	96.21	610.28	
2	0	134.60	415.45	
3	0	361.66	1599.51	
4	0	22.14	289.54	

	Customer_Status	Churn_Category	Churn_Reason
0	Stayed	NaN	NaN
1	Stayed	NaN	NaN
2	Churned	Competitor	Competitor had better devices
3	Churned	Dissatisfaction	Product dissatisfaction
4	Churned	Dissatisfaction	Network reliability

[5 rows x 32 columns]

```
In [2]: # Drop Unwanted Columns
data = data.drop(
    ['Customer_ID', 'Churn_Category', 'Churn_Reason'],
    axis=1,
    errors='ignore'
)

# Fill Missing Values
data = data.fillna('Unknown')

# Columns To Encode
columns_to_encode = [

    'Gender',
    'Married',
    'State',
    'Value_Deal',
    'Phone_Service',
    'Multiple_Lines',
```

```

    'Internet_Service',
    'Internet_Type',
    'Online_Security',
    'Online_Backup',
    'Device_Protection_Plan',
    'Premium_Support',
    'Streaming_TV',
    'Streaming_Movies',
    'Streaming_Music',
    'Unlimited_Data',
    'Contract',
    'Paperless_Billing',
    'Payment_Method'

]

# Encode Data
label_encoders = {}

for column in columns_to_encode:

    label_encoders[column] = LabelEncoder()

    data[column] = label_encoders[column].fit_transform(
        data[column].astype(str)
    )

# Encode Target Column
data['Customer_Status'] = data['Customer_Status'].map({
    'Stayed': 0,
    'Churned': 1
})

print(data.head())

```

	Gender	Age	Married	State	Number_of_Referrals	Tenure_in_Months	\
0	1	35	0	4	7	27	
1	0	45	1	12	14	13	
2	1	51	0	21	4	35	
3	1	79	0	9	3	21	
4	0	80	0	17	3	8	

	Value_Deal	Phone_Service	Multiple_Lines	Internet_Service	...	Contract
0	5	1	0	1	...	
1	5	1	2	1	...	
2	4	1	0	1	...	
3	3	1	0	1	...	
4	5	1	0	1	...	

	Paperless_Billing	Payment_Method	Monthly_Charge	Total_Charges	\
0	1	1	65.6	593.30	
1	0	1	-4.0	542.40	
2	1	0	73.9	280.85	
3	1	0	98.0	1237.85	
4	1	1	83.9	267.40	

	Total_Refunds	Total_Extra_Data_Charges	Total_Long_Distance_Charges	\
0	0.00	0	381.51	
1	38.33	10	96.21	
2	0.00	0	134.60	
3	0.00	0	361.66	
4	0.00	0	22.14	

	Total_Revenue	Customer_Status
0	974.81	0
1	610.28	0
2	415.45	1
3	1599.51	1
4	289.54	1

[5 rows x 29 columns]

```
In [3]: # Features
X = data.drop('Customer_Status', axis=1)

# Target
y = data['Customer_Status']

# Split Data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
print("Data Split Successfully")
```

Data Split Successfully

```
In [4]: # Initialize Model
rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

# Train Model
rf_model.fit(X_train, y_train)

print("Model Trained Successfully")
```

Model Trained Successfully

```
In [5]: # Predictions
y_pred = rf_model.predict(X_test)

# Confusion Matrix
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Confusion Matrix:

```
[[789  52]
 [135 226]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.94	0.89	841
1	0.81	0.63	0.71	361
accuracy			0.84	1202
macro avg	0.83	0.78	0.80	1202
weighted avg	0.84	0.84	0.84	1202

```
In [6]: # Feature Importance
importances = rf_model.feature_importances_

indices = np.argsort(importances)[::-1]

# Plot Graph
plt.figure(figsize=(15,6))

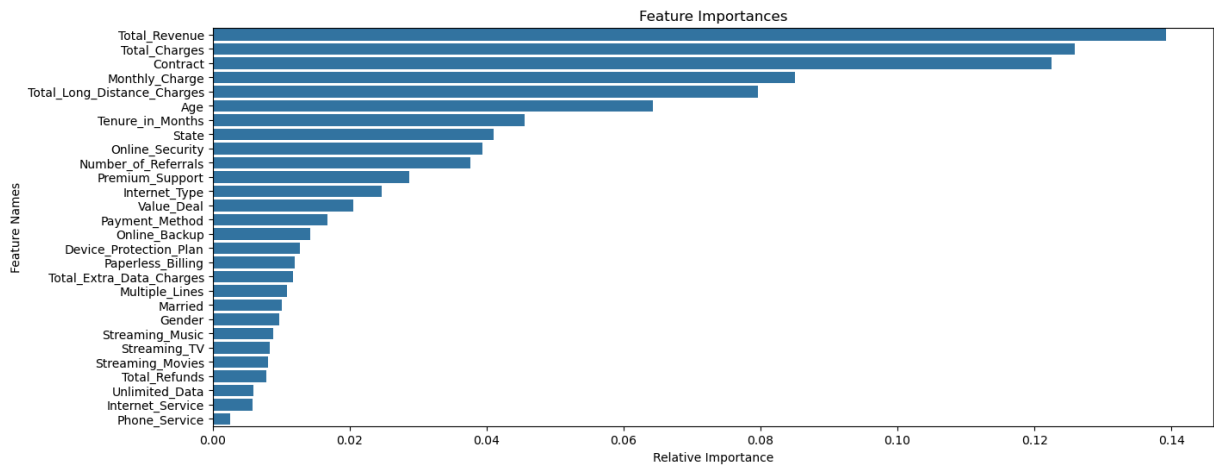
sns.barplot(
    x=importances[indices],
    y=X.columns[indices]
)
```

```
plt.title('Feature Importances')

plt.xlabel('Relative Importance')

plt.ylabel('Feature Names')

plt.show()
```



```
In [7]: # Joiner File Path
join_file_path = r"D:\Data\Customer_Status_Predicted.xlsx"

# Read Joiner Data
new_data = pd.read_excel(
    join_file_path,
    sheet_name='vw_joindata'
)

# Keep Original Copy
original_data = new_data.copy()

# Clean Column Names
new_data.columns = new_data.columns.str.strip() \
    .str.replace(" ", "_") \
    .str.replace(r"^[^w]", "", regex=True)

# Show Data
print(new_data.head())
```

	Customer_ID	Gender	Age	Married	State	Number_of_Referrals	\
0	93520-GUJ	Female	67	No	Gujarat	13	
1	57256-BIH	Female	18	No	Bihar	9	
2	72357-MAD	Female	53	No	Madhya Pradesh	14	
3	66612-KAR	Female	58	Yes	Karnataka	11	
4	22119-WES	Male	31	Yes	West Bengal	5	

	Tenure_in_Months	Value_Deal	Phone_Service	Multiple_Lines	...	\
0	19	Deal 5	Yes	Yes	...	
1	7	NaN	Yes	No	...	
2	12	Deal 5	Yes	No	...	
3	18	NaN	Yes	No	...	
4	5	NaN	Yes	No	...	

	Payment_Method	Monthly_Charge	Total_Charges	Total_Refunds	\
0	Bank Withdrawal	72.10	72.1	0.0	
1	Credit Card	19.85	57.2	0.0	
2	Credit Card	44.30	44.3	0.0	
3	Credit Card	19.95	58.0	0.0	
4	Credit Card	20.05	33.7	0.0	

	Total_Extra_Data_Charges	Total_Long_Distance_Charges	Total_Revenue	\
0	0		7.77	79.87
1	0		9.36	66.56
2	0		42.95	87.25
3	0		8.07	66.07
4	0		3.62	37.32

	Customer_Status	Churn_Category	Churn_Reason
0	Joined	NaN	NaN
1	Joined	NaN	NaN
2	Joined	NaN	NaN
3	Joined	NaN	NaN
4	Joined	NaN	NaN

[5 rows x 32 columns]

```
In [8]: # Drop Extra Columns
new_data = new_data.drop(
    ['Customer_ID', 'Customer_Status', 'Churn_Category', 'Churn_Reason'],
    axis=1,
    errors='ignore'
)

# Fill Missing Values
new_data = new_data.fillna('Unknown')

# Encode Data
for column in new_data.columns:

    if column in label_encoders:

        new_data[column] = label_encoders[column].transform(
            new_data[column].astype(str)
        )
```

```
# Match Training Columns
new_data = new_data[X.columns]

print("Joiner Data Ready")
```

Joiner Data Ready

```
In [9]: # Predict
new_predictions = rf_model.predict(new_data)

# Add Prediction Column
original_data['Customer_Status_Predicted'] = new_predictions

# Keep Only Churned Customers
original_data = original_data[
    original_data['Customer_Status_Predicted'] == 1
]

# Show Output
print(original_data.head())

# Total Predicted Customers
print(
    "Total Predicted Customers:",
    len(original_data)
)

# Save Final Output
original_data.to_csv(
    r"D:\Data\Predictions.csv",
    index=False
)

print("Prediction File Saved Successfully")
```


	Customer_ID	Gender	Age	Married	State	Number_of_Referrals	\
0	93520-GUJ	Female	67	No	Gujarat	13	
1	57256-BIH	Female	18	No	Bihar	9	
2	72357-MAD	Female	53	No	Madhya Pradesh	14	
3	66612-KAR	Female	58	Yes	Karnataka	11	
4	22119-WES	Male	31	Yes	West Bengal	5	

	Tenure_in_Months	Value_Deal	Phone_Service	Multiple_Lines	...	\
0	19	Deal 5	Yes	Yes	...	
1	7	NaN	Yes	No	...	
2	12	Deal 5	Yes	No	...	
3	18	NaN	Yes	No	...	
4	5	NaN	Yes	No	...	

	Monthly_Charge	Total_Charges	Total_Refunds	Total_Extra_Data_Charges	\
0	72.10	72.1	0.0	0	
1	19.85	57.2	0.0	0	
2	44.30	44.3	0.0	0	
3	19.95	58.0	0.0	0	
4	20.05	33.7	0.0	0	

	Total_Long_Distance_Charges	Total_Revenue	Customer_Status	Churn_Category	\
0	7.77	79.87	Joined	NaN	
1	9.36	66.56	Joined	NaN	
2	42.95	87.25	Joined	NaN	
3	8.07	66.07	Joined	NaN	
4	3.62	37.32	Joined	NaN	

	Churn_Reason	Customer_Status_Predicted
0	NaN	1
1	NaN	1
2	NaN	1
3	NaN	1
4	NaN	1

[5 rows x 33 columns]

Total Predicted Customers: 385

Prediction File Saved Successfully

In []: